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EDUCATION

B.S. Aerospace Engineering, California State University, Long Beach (CSULB) Expected May 2028

GPA: 3.55 | **Coursework:** Engineering Design Graphics, Mechanics of Materials, Fluid Mechanics, Flight Mechanics, Statics, Dynamics

EXPERIENCE

Aerostructures Assistant Lead, Beach Launch Team – *CSULB* June 2026 – Present

- **Pressure Vessel Testing & DAQ:** Manually operated high-pressure pumping systems to conduct a 4500 psi hydrostatic proof testing on a Nitrogen tank COPV, logging real-time data to validate structural safety margins.
- **Bulkhead Trade Study & Layout Optimization:** Performed hand-calculated stress-strain trade studies comparing two bulkhead designs; selected a lower-strength design that satisfied factor-of-safety constraints while critical space was preserved for fluids and avionics routing.
- **Mass Reduction of Bulkhead:** Calculated bulkhead load paths and allowable material removal using factor-of-safety constraints, reducing bulkhead dry mass by 2% —approximately [2.75 lbs] while maintaining structural requirements.
- **Quality Assurance & CAD Audit:** Designed 8 bulkheads for manufacturability using available resources and ensured compliance with subsystem space requirements.

Aerostructures Engineer, Beach Launch Team – *CSULB* August 2025 – June 2026

- **Pre-Test Integration & Test Operations:** Participated in aerostructures pre-test checks and structural vehicle integration, contributing to a successful static fire test.
- **Composite Layup & Fin Fabrication:** Machined precision leading-edge bevels on rocket fins and executed tip-to-tip carbon-fiber wet layups, optimizing torsional and structural stiffness.
- **Precision Sheet Metal Fabrication:** Fabricated an aluminum sheet metal airframe skin bay, precision-drilling and deburring mounting holes to ensure exact alignment with internal structural frames.

Project Experience – *California State University, Long Beach*

- **Butterfly Valve Design (December 2025):** Modeled a complete butterfly valve assembly in CATIA as a personal design project, detailing multi-part component fitment and kinematic movement.
- **L1 Solid Rocket Design and Build (September 2026-present):** Designed a rocket to fly a commercial H-class motor. Manufactured an airframe using fiberglass wet-lay-ups and precision wood work.

SOFTWARE AND SKILLS

- **Software & CAD/CAM:** CATIA, Onshape, MATLAB, Python, Ansys FEA, Technical Drawings, Microsoft Excel
- **Manufacturing & Fabrication:** Precision sheet metal fabrication, composite material wet layups, machining (Lathe, Drill Press), deburring, structural assembly
- **Testing & Operations:** Hydrostatic proof testing (up to 4,500 psi), sensor data acquisition, COPV validation, pre-test structural integration, launch pad operations
- **Aerostructures & Analysis:** Longitudinal airframe design, structural load distribution, fuselage rigidity optimization.